POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6023B | Semester VI | Theory | 4 Credits

Additive Manufacturing

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6023B as Additive Manufacturing in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on the NPTEL IISc Bangalore course Advances in Additive Manufacturing of Materials and a peer-reviewed open-access review of AM process classification. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Powder, resin, solvent, laser/beam, hot-end, moving-axis and post-processing work requires machine-specific training, authorised ventilation/containment, PPE, fire controls and documented waste practices.

Verified source note: Verified sources support layer-by-layer CAD/scanning/slicing workflow, ISO/ASTM process categories, material/process selection, binder jetting, PBF/DED, material extrusion, quality limitations, post-processing, applications and emerging data/AI context. The four modules are a learner-oriented sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Additive Manufacturing studies the layer-by-layer creation of components from digital data. It connects design, materials, process families, data preparation, build strategy, quality, post-processing and application choice so that learners can assess where additive methods complement conventional production.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Engineering graphics/CAD awareness, manufacturing processes, engineering materials, basic measurement and strict workshop/laboratory safety.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain additive-manufacturing principles, digital workflow and standard process classification.
2. Explain major additive-process families, feedstocks, equipment elements and process selection.
3. Explain design-for-additive-manufacturing, build preparation, supports and post-processing concepts.
4. Explain quality, defects, testing, applications, sustainability and emerging developments in additive manufacturing.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain additive-manufacturing concepts, digital workflow and process classifications.	Understand
CO2	Explain major AM process families, materials, equipment and selection criteria.	Understand
CO3	Apply design/build-preparation and post-processing concepts to a simple part case.	Apply
CO4	Explain AM quality challenges, applications, sustainability and emerging technology directions.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	AM Fundamentals and Digital Workflow	Definition and value proposition; additive versus subtractive/formative methods; CAD or scan input, file preparation, slicing, layer-by-layer build, inspection and post-processing; standard process categories.	Approx. 14 hours	CO1	Module 1
2	Processes, Materials and Equipment	Material extrusion, vat photopolymerisation, binder/material jetting, powder-bed fusion, directed-energy deposition and sheet-lamination concepts; feedstock forms, machine subsystems and application fit.	Approx. 16 hours	CO2	Module 2
3	Design, Build Preparation and Post-Processing	Design for AM, orientation, supports, nesting/build strategy, parameter awareness, removal/cleaning/curing/heat treatment/finishing and inspection planning.	Approx. 16 hours	CO3	Module 3
4	Quality, Applications and Responsible Development	Defects, anisotropy, porosity, dimensional/surface challenges, inspection and traceability; applications; sustainability and circularity; data/AI and emerging manufacturing opportunities.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

AM Fundamentals and Digital Workflow

Definition and value proposition; additive versus subtractive/formative methods; CAD or scan input, file preparation, slicing, layer-by-layer build, inspection and post-processing; standard process categories.

CO1 · Approx. 14 hours

Processes, Materials and Equipment

Material extrusion, vat photopolymerisation, binder/material jetting, powder-bed fusion, directed-energy deposition and sheet-lamination concepts; feedstock forms, machine subsystems and application fit.

CO2 · Approx. 16 hours

Design, Build Preparation and Post-Processing

Design for AM, orientation, supports, nesting/build strategy, parameter awareness, removal/cleaning/curing/heat treatment/finishing and inspection planning.

CO3 · Approx. 16 hours

Quality, Applications and Responsible Development

Defects, anisotropy, porosity, dimensional/surface challenges, inspection and traceability; applications; sustainability and circularity; data/AI and emerging manufacturing opportunities.

CO4 · Approx. 14 hours

MODULE 1

AM Fundamentals and Digital Workflow

Definition and value proposition; additive versus subtractive/formative methods; CAD or scan input, file preparation, slicing, layer-by-layer build, inspection and post-processing; standard process categories.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

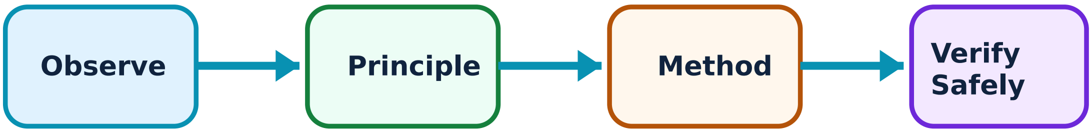
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for AM Fundamentals and Digital Workflow: identify the system, state its principle, follow the correct method and verify the result safely.

1. What makes manufacturing additive?−

Additive manufacturing creates a part by selectively joining or depositing material layer by layer from digital geometry. It can support complex geometry, customisation, rapid iteration and on-demand production, but it also has constraints in size, rate, finish, repeatability, material and cost.

Study and application method

Compare an additive and a conventional route for one part, stating geometry, volume, material, tooling, lead time, finish, tolerance, waste and post-processing assumptions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare an additive and a conventional route for one part, stating geometry, volume, material, tooling, lead time, finish, tolerance, waste and post-processing assumptions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not assume AM is automatically lower cost or lower impact. The appropriate route depends on the full product, production and life-cycle context.

2. Digital build workflow and classification−

A typical workflow moves from CAD or scan data through geometry checking, orientation, slicing and parameter setup to build, removal, finishing and inspection. Widely used ISO/ASTM categories include vat photopolymerisation, material jetting, binder jetting, material extrusion, powder-bed fusion, directed-energy deposition and sheet lamination.

Study and application method

Create a process map for a simple component and identify where geometry error, incorrect orientation, support need, data conversion, parameter selection or post-build distortion may enter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Never send unverified data or parameters to a live machine. Approved files, access control and simulation/check procedures are part of safe production.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Processes, Materials and Equipment

Material extrusion, vat photopolymerisation, binder/material jetting, powder-bed fusion, directed-energy deposition and sheet-lamination concepts; feedstock forms, machine subsystems and application fit.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

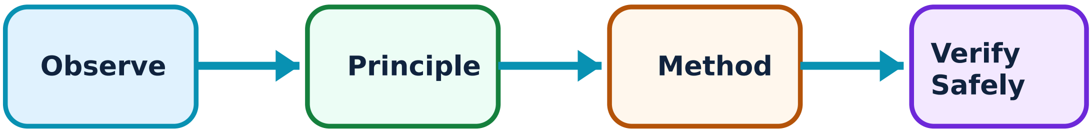
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Processes, Materials and Equipment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Process families and energy/material mechanisms−

AM families differ by feedstock and joining mechanism: extrusion deposits material through a nozzle; vat processes cure photosensitive material; jetting deposits droplets/binder; PBF selectively fuses powder; DED adds material into a focused energy zone; sheet lamination joins sheets. Each has distinct capability and constraints.

Study and application method

Build a comparison table for three process families: feedstock, energy/joining mechanism, typical geometry/support characteristics, likely materials, strengths, limitations and example application.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a comparison table for three process families: feedstock, energy/joining mechanism, typical geometry/support characteristics, likely materials, strengths, limitations and example application.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Avoid equating marketing names with universal capability. Machine model, material qualification, parameter set and post-processing determine actual results.

2. Materials and machine subsystems−

Polymers, metals, ceramics, composites, binders and soft/biological materials may be used in AM, subject to process compatibility. Systems can include motion stages, feed/bed handling, energy source, atmosphere control, sensors, software and extraction/filtration.

Study and application method

For a stated material/process combination, identify feedstock form, critical material property, equipment subsystem, expected risk to quality and an appropriate source of validated material data.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a stated material/process combination, identify feedstock form, critical material property, equipment subsystem, expected risk to quality and an appropriate source of validated material data.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** Powder and resin handling may expose workers to inhalation, skin sensitisation, fire or contamination risks. Follow the material safety data and authorised controls—not generic internet advice.

> **Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Design, Build Preparation and Post-Processing

Design for AM, orientation, supports, nesting/build strategy, parameter awareness, removal/cleaning/curing/heat treatment/finishing and inspection planning.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

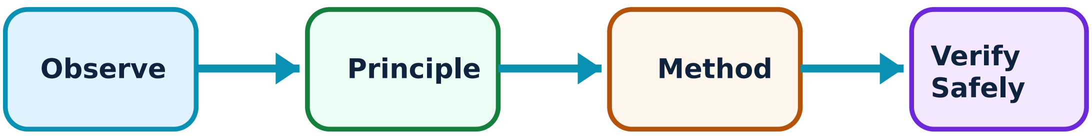
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Design, Build Preparation and Post-Processing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Design for additive manufacturing−

DfAM uses the design freedom of AM while respecting process limits such as minimum feature size, unsupported overhangs, trapped material, anisotropy, build volume, thermal distortion and achievable finish. Orientation influences support, time, strength direction and surface quality.

Study and application method

Review a simple CAD concept and list features to retain, modify or verify: wall thickness, channels, overhangs, powder/resin escape, build orientation, support removal and measurement access.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Review a simple CAD concept and list features to retain, modify or verify: wall thickness, channels, overhangs, powder/resin escape, build orientation, support removal and measurement access.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A classroom DfAM checklist is not a production release. Critical parts require validated design rules, process qualification and independent inspection planning.

2. Post-processing and readiness for use−

Many AM routes need support removal, cleaning, curing, debinding, sintering, heat treatment, machining, polishing or coating to achieve required function and quality. The build itself may establish geometry but not final material properties or finish.

Study and application method

Prepare a post-processing flow that states the purpose of each stage, handling precautions, evidence of completion and inspection point before the part is released.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a post-processing flow that states the purpose of each stage, handling precautions, evidence of completion and inspection point before the part is released.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Post-processing can introduce heat, chemicals, dust, sharp edges and residual stresses. Do not bypass controls or assume a visually acceptable part meets performance requirements.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Quality, Applications and Responsible Development

Defects, anisotropy, porosity, dimensional/surface challenges, inspection and traceability; applications; sustainability and circularity; data/AI and emerging manufacturing opportunities.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

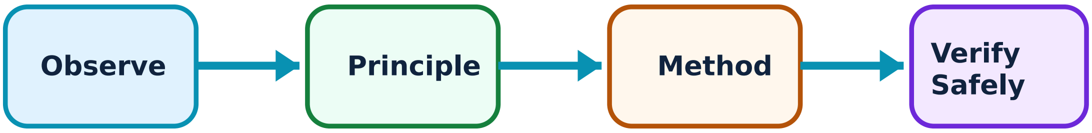
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Quality, Applications and Responsible Development: identify the system, state its principle, follow the correct method and verify the result safely.

1. Quality, defects and verification−

AM quality can be affected by layer bonding, voids/porosity, thermal history, residual stress, distortion, incomplete curing/fusion, surface stair-stepping, powder/resin contamination and parameter variation. Verification must connect requirements to material, geometry, surface and process evidence.

Study and application method

For a part requirement, construct a quality plan with critical characteristics, likely defect, inspection approach, data to retain, acceptance authority and escalation path.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a part requirement, construct a quality plan with critical characteristics, likely defect, inspection approach, data to retain, acceptance authority and escalation path.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Non-destructive indications, test coupons and visual checks have limits. Critical applications must follow applicable qualification, certification and inspection requirements.

2. Applications, sustainability and emerging directions−

AM is used for prototypes, tooling, customised medical/aerospace/automotive components, complex thermal-fluid structures and local/on-demand production. Sustainability depends on energy, material yield, support/waste, transport, repair/reuse and end-of-life. Data analytics and AI can support process monitoring and quality prediction when responsibly validated.

Study and application method

Evaluate an application with a life-cycle table: functional need, production quantity, material yield, energy, logistics, support/waste, repairability, recyclability, quality burden and data requirements.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Evaluate an application with a life-cycle table: functional need, production quantity, material yield, energy, logistics, support/waste, repairability, recyclability, quality burden and data requirements.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not make sustainability or performance claims from one benefit alone. Account for energy, post-processing, failures, material recovery and validation burden.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: AM Fundamentals and Digital Workflow

Use the concept flow to revise observation, principle, method and verification.

Module 2: Processes, Materials and Equipment

Use the concept flow to revise observation, principle, method and verification.

Module 3: Design, Build Preparation and Post-Processing

Use the concept flow to revise observation, principle, method and verification.

Module 4: Quality, Applications and Responsible Development

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Map the CAD-to-part workflow for an example component and identify quality checkpoints.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare three AM process families for a component using a materials–geometry–quality–cost table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Perform a DfAM review of a simple part model, including orientation, supports, internal channels and post-processing access.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a simulated post-processing and inspection plan, separating mandatory checks from assumptions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Evaluate an AM application case using technical, environmental, economic and safety criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — AM Fundamentals and Digital Workflow

Explain What makes manufacturing additive? and state one correct method, application or precaution.−

Additive manufacturing creates a part by selectively joining or depositing material layer by layer from digital geometry. It can support complex geometry, customisation, rapid iteration and on-demand production, but it also has constraints in size, rate, finish, repeatability, material and cost.

Answer framework: Compare an additive and a conventional route for one part, stating geometry, volume, material, tooling, lead time, finish, tolerance, waste and post-processing assumptions.

Important point: Do not assume AM is automatically lower cost or lower impact. The appropriate route depends on the full product, production and life-cycle context.

Explain Digital build workflow and classification and state one correct method, application or precaution.−

A typical workflow moves from CAD or scan data through geometry checking, orientation, slicing and parameter setup to build, removal, finishing and inspection. Widely used ISO/ASTM categories include vat photopolymerisation, material jetting, binder jetting, material extrusion, powder-bed fusion, directed-energy deposition and sheet lamination.

Answer framework: Create a process map for a simple component and identify where geometry error, incorrect orientation, support need, data conversion, parameter selection or post-build distortion may enter.

Important point: Never send unverified data or parameters to a live machine. Approved files, access control and simulation/check procedures are part of safe production.

Module 2 — Processes, Materials and Equipment

Explain Process families and energy/material mechanisms and state one correct method, application or precaution.−

AM families differ by feedstock and joining mechanism: extrusion deposits material through a nozzle; vat processes cure photosensitive material; jetting deposits droplets/binder; PBF selectively fuses powder; DED adds material into a focused energy zone; sheet lamination joins sheets. Each has distinct capability and constraints.

Answer framework: Build a comparison table for three process families: feedstock, energy/joining mechanism, typical geometry/support characteristics, likely materials, strengths, limitations and example application.

Important point: Avoid equating marketing names with universal capability. Machine model, material qualification, parameter set and post-processing determine actual results.

Explain Materials and machine subsystems and state one correct method, application or precaution.−

Polymers, metals, ceramics, composites, binders and soft/biological materials may be used in AM, subject to process compatibility. Systems can include motion stages, feed/bed handling, energy source, atmosphere control, sensors, software and extraction/filtration.

Answer framework: For a stated material/process combination, identify feedstock form, critical material property, equipment subsystem, expected risk to quality and an appropriate source of validated material data.

Important point: Powder and resin handling may expose workers to inhalation, skin sensitisation, fire or contamination risks. Follow the material safety data and authorised controls —not generic internet advice.

Module 3 — Design, Build Preparation and Post-Processing

Explain Design for additive manufacturing and state one correct method, application or precaution.−

DfAM uses the design freedom of AM while respecting process limits such as minimum feature size, unsupported overhangs, trapped material, anisotropy, build volume, thermal distortion and achievable finish. Orientation influences support, time, strength direction and surface quality.

Answer framework: Review a simple CAD concept and list features to retain, modify or verify: wall thickness, channels, overhangs, powder/resin escape, build orientation, support removal and measurement access.

Important point: A classroom DfAM checklist is not a production release. Critical parts require validated design rules, process qualification and independent inspection planning.

Explain Post-processing and readiness for use and state one correct method, application or precaution.−

Many AM routes need support removal, cleaning, curing, debinding, sintering, heat treatment, machining, polishing or coating to achieve required function and quality. The build itself may establish geometry but not final material properties or finish.

Answer framework: Prepare a post-processing flow that states the purpose of each stage, handling precautions, evidence of completion and inspection point before the part is released.

Important point: Post-processing can introduce heat, chemicals, dust, sharp edges and residual stresses. Do not bypass controls or assume a visually acceptable part meets performance requirements.

Module 4 — Quality, Applications and Responsible Development

Explain Quality, defects and verification and state one correct method, application or precaution.−

AM quality can be affected by layer bonding, voids/porosity, thermal history, residual stress, distortion, incomplete curing/fusion, surface stair-stepping, powder/resin contamination and parameter variation. Verification must connect requirements to material, geometry, surface and process evidence.

Answer framework: For a part requirement, construct a quality plan with critical characteristics, likely defect, inspection approach, data to retain, acceptance authority and escalation path.

Important point: Non-destructive indications, test coupons and visual checks have limits. Critical applications must follow applicable qualification, certification and inspection requirements.

Explain Applications, sustainability and emerging directions and state one correct method, application or precaution.−

AM is used for prototypes, tooling, customised medical/aerospace/automotive components, complex thermal-fluid structures and local/on-demand production. Sustainability depends on energy, material yield, support/waste, transport, repair/reuse and end-of-life. Data analytics and AI can support process monitoring and quality prediction when responsibly validated.

Answer framework: Evaluate an application with a life-cycle table: functional need, production quantity, material yield, energy, logistics, support/waste, repairability, recyclability, quality burden and data requirements.

Important point: Do not make sustainability or performance claims from one benefit alone. Account for energy, post-processing, failures, material recovery and validation burden.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: AM Fundamentals and Digital Workflow.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Processes, Materials and Equipment.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Design, Build Preparation and Post-Processing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Quality, Applications and Responsible Development.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and value proposition; additive versus subtractive/formative methods; CAD or scan input, file preparation, slicing, layer-by-layer build, inspection and post-processing; standard process categories..
2. Develop a complete answer or practical plan for: Material extrusion, vat photopolymerisation, binder/material jetting, powder-bed fusion, directed-energy deposition and sheet-lamination concepts; feedstock forms, machine subsystems and application fit..
3. Develop a complete answer or practical plan for: Design for AM, orientation, supports, nesting/build strategy, parameter awareness, removal/cleaning/curing/heat treatment/finishing and inspection planning..
4. Develop a complete answer or practical plan for: Defects, anisotropy, porosity, dimensional/surface challenges, inspection and traceability; applications; sustainability and circularity; data/AI and emerging manufacturing opportunities..

LAST-MILE REVISION

Quick Revision

Module 1: AM Fundamentals and Digital Workflow

Definition and value proposition; additive versus subtractive/formative methods; CAD or scan input, file preparation, slicing, layer-by-layer build, inspection and post-processing; standard process categories.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Processes, Materials and Equipment

Material extrusion, vat photopolymerisation, binder/material jetting, powder-bed fusion, directed-energy deposition and sheet-lamination concepts; feedstock forms, machine subsystems and application fit.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Design, Build Preparation and Post-Processing

Design for AM, orientation, supports, nesting/build strategy, parameter awareness, removal/cleaning/curing/heat treatment/finishing and inspection planning.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Quality, Applications and Responsible Development

Defects, anisotropy, porosity, dimensional/surface challenges, inspection and traceability; applications; sustainability and circularity; data/AI and emerging manufacturing opportunities.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
What makes manufacturing additive?	Additive manufacturing creates a part by selectively joining or depositing material layer by layer from digital geometry.
Digital build workflow and classification	A typical workflow moves from CAD or scan data through geometry checking, orientation, slicing and parameter setup to build, removal, finishing and inspection.
Process families and energy/material mechanisms	AM families differ by feedstock and joining mechanism: extrusion deposits material through a nozzle; vat processes cure photosensitive material; jetting deposits droplets/binder; PBF selectively fuses powder; DED adds material into a focused energy zone; sheet lamination joins sheets.
Materials and machine subsystems	Polymers, metals, ceramics, composites, binders and soft/biological materials may be used in AM, subject to process compatibility.
Design for additive manufacturing	DfAM uses the design freedom of AM while respecting process limits such as minimum feature size, unsupported overhangs, trapped material, anisotropy, build volume, thermal distortion and achievable finish.
Post-processing and readiness for use	Many AM routes need support removal, cleaning, curing, debinding, sintering, heat treatment, machining, polishing or coating to achieve required function and quality.
Quality, defects and verification	AM quality can be affected by layer bonding, voids/porosity, thermal history, residual stress, distortion, incomplete curing/fusion, surface stair-stepping, powder/resin contamination and parameter variation.
Applications, sustainability and emerging directions	AM is used for prototypes, tooling, customised medical/aerospace/automotive components, complex thermal-fluid structures and local/on-demand production.

LEARNING RESOURCES

References

Recommended additive-manufacturing references

1. I. Gibson, D. Rosen and B. Stucker, Additive Manufacturing Technologies.
2. A. Gebhardt and J.-S. Hötter, Additive Manufacturing: 3D Printing for Prototyping and Manufacturing.
3. T. D. Wohlers et al., Wohlers Report (consult current edition for industry data).
4. ASTM/ISO 52900 terminology and principles (consult the current authorised standard).

Approved public additive-manufacturing sources

- https://onlinecourses.nptel.ac.in/noc25_mm02/preview
- https://pmc.ncbi.nlm.nih.gov/articles/PMC10880673/

These sources provide the verified study basis while official Course 6023B detail is unavailable; they do not replace equipment manuals, material safety data, applicable standards or authorised process specifications.